QMS
12 - Std
Time : 3.00 hrs Salem

QUARTERLY EXAMINATION - 2026
CHEMISTRY

Marks : 70

Note: i) Answer all the questions. ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer. (15 x 1 = 15)

1. The incorrect statement among the following is
   a) Nickel is refined by Mond's process b) Titanium is refined by Van Arkel's process
   c) Zinc blende is concentrated by froth floatation
   d) In the metallurgy of gold, the metal is leached with dilute sodium chloride solution.
2. XeF_6 on complete hydrolysis produces
   a) $XeOF_4$ b) XeO_2F_2 c) XeO_3 d) XeO_2
3. Which of the following is not SP² hybridised?
   a) Graphite b) Graphene c) Fullerene d) Dry ice
4. In acid medium potassium permanganate oxidizes oxalic acid to
   a) Oxalate b) Carbon di oxide c) Acetate d) Acetic acid
5. Graphite and diamonds are
   a) both Covalent crystals b) Covalent and molecular crystals
   c) Ionic and Covalent crystals d) both molecular crystals
6. The rate constant of a reaction is $5.8 \times 10^{-2}\ s^{-1}$. The order of the reaction is
   a) First order b) Zero order c) Second order d) Third order
7. Which of the following can act as Lowery-Bronsted acid as well as base?
   a) HCl b) SO_4^{2-} c) HPO_4^{2-} d) Br^-
8. When phenol is treated with $CHCl_3$ and NaOH, the product formed is
   a) Salicylic acid b) Benzoic acid c) Salicylaldehyde d) Benzaldehyde
9. In which of the following reactions new carbon-carbon bond is not formed?
   a) Aldol Condensation b) Friedal Craft reaction
   c) Kolbe's reaction d) Wolf Kishner reaction
10. The process of converting hydrated alumina into anhydrous alumina is called
    a) Roasting b) Smelting
    c) Auto reduction d) Calcination
11. The fraction of total volume occupied by the atoms in a simple cubic is
    a) $\left(\frac{\pi}{4\sqrt{2}}\right)$ b) $\left(\frac{\pi}{6}\right)$ c) $\left(\frac{\pi}{4}\right)$ d) $\left(\frac{\pi}{3\sqrt{2}}\right)$
12. Sodium salt of tetraboric acid is known as
    a) B_2H_6 b) $Na_2B_4O_7.10H_2O$ c) Na_2BO_3 d) H_3BO_3
13. Isopropyl benzene on air oxidation in the presence of dilute acid gives
    a) C_6H_5COOH b) $C_6H_5COCH_3$ c) $C_6H_5COC_6H_5$ d) C_6H_5OH
14. Dry distillation of calcium formate gives
    a) Ethanal b) Ethanol c) Methanal d) Propanone

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15. Among the following, which will not be hydrolysed?
    a) Sodium chloride b) Sodium formate
    c) Ammonium formate d) Ammonium Nitrate

II Note: Answer any Six Questions. Question number 24 is Compulsory. (6 x 2 = 12)

16. Write Mond's process of refining Nickel.
17. Write the uses of Boric acid.
18. How bleaching powder is prepared?
19. Which is more stable? Fe^{3+} or Fe^{2+} - Explain.
20. Why ionic crystals are hard and brittle?
21. Define Solubility product.
22. Write the dehydration reaction of Glycerol.
23. How pinacol is prepared?
24. The rate constant for a first order reaction is $1.54 \times 10^{-3} s^{-1}$. Calculate its Half Life time.

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III Note: Answer any six questions. Question number 33 is Compulsory. (6 x 3 = 18)

25. What is urotrophine? How it is prepared? Give the structure.
26. Explain acid leaching with an example.
27. Write the conditions for catanation.
28. Write the classifications of solids.
29. Derive an equation for half life period of first order reaction.
30. Write the reason for the anomolus behaviour of Nitrogen.
31. Give the structure and IUPAC name of metamers of 2-methoxy propane.
32. Write the properties of Interstitial Compounds.
33. The Ka value for HCN is 10^{-9}. What is the pH of 0.4M HCN Solution.

IV Note: Answer all the Questions. (5 x 5 = 25)

34. a) Explain froth flotation process. ***(OR)***
    b) i) How will you convert boric acid to boron nitride? ii) Give the uses of Silicones.
35. a) i) What are interhalogen compounds?
    ii) Write the molecular formula and draw the structure of sulphurous acid and Marshall's acid. ***(OR)***
    b) Derive an expression for Oswald's dilution Law.
36. a) Explain Schottky defect and Frenkel defect with suitable example. ***(OR)***
    b) i) Write Schotten-Boumann Reaction.
    ii) Write the Lucas test to distinguish 1°, 2° and 3° alcohols.
37. a) Compare Lanthanides and Actinides. ***(OR)***
    b) Describe the structure of Diborane.
38. a) i) Write the rate law for the following reactions.
    1) A reaction that is $3/2$, order in x and zero order in y.
    2) A reaction that is second order in NO and first order in Br_2.
    ii) Give three examples for first order reactions. ***(OR)***
    b) A compound (A) with molecular formula C_2H_3N on acid hydrolysis gives (B) which reacts with thionyl chloride to give compound (C). Benzene reacts with compound (C) in the presence of an hydrous $AlCl_3$ to give compound (D). Identify A, B, C, D and write the reactions.

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